MSc Data Science
DS601: Mathematical Foundations for Data Science
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Tutorial - 6

Question 1 — Joint PMF

Two servers in a data center intermittently fail during peak load. Let X = number of failures at Server A and Y = number of failures at Server B during a given hour, where $X, Y \in \{0, 1, 2\}$. Suppose the joint PMF has the form

$$p(x, y) = c(x + y + 1), \qquad x, y \in \{0, 1, 2\}.$$

(a) Find the value of c that makes this a valid joint PMF.

(b) Compute $P(X + Y \leq 2)$.

(c) Without computing the full marginal distributions of X and Y in advance, determine whether X and Y are independent. Justify your conclusion using an explicit numerical check.

Question 2 — Joint PDF

Let X and Y be continuous random variables with joint density

$$f(x, y) = c\,xy, \qquad 0 < y < x < 1,$$

and $f(x, y) = 0$ otherwise.

(a) Find the value of c.

(b) Compute $P(Y > X/2)$. Be careful to set up the integration limits correctly using the given support.

(c) Based only on the shape of the support region (not on computing the marginals), explain why X and Y cannot be independent.

Question 3 — Marginal Distributions

A joint density models two proportions X and Y (e.g., fractions of two related resources consumed) with

$$f(x, y) = 3(x + y), \qquad x > 0, y > 0, x + y < 1,$$

and $f(x, y) = 0$ otherwise.

(a) Derive the marginal PDF $f_X(x)$ of X, being careful that the limits of integration used to eliminate y depend on x. State its support.

(b) Using $f_X(x)$, compute $P(X > 0.5)$.

(c) Using the marginal PDFs $f_X(x)$ and $f_Y(y)$ (the latter by symmetry), determine whether X and Y are independent. Justify your answer with a specific comparison.

Dataset (used throughout Q4, Q5, Q6)

Top speed (km/h) recorded for 8 bikes in each category:

Observation	Sport Bike	Cruiser	Off-road Bike
1	170	168	100
2	175	172	105
3	180	176	110
4	185	180	115
5	190	184	120
6	178	170	108
7	182	178	112
8	176	174	106

All three categories are assumed to follow a **Normal distribution**, each with its own mean and variance. Use the **population convention** (divide by n, not $n - 1$) throughout.

Question 4 — Moments and Probability Density Functions

Using the dataset above, calculate for **each category** (Sport, Cruiser, Off-road):

1. The first raw moment:

$$\mu_1' = E[X] = (1/n) \Sigma_{i=1}^{n} x_i$$

2. The second raw moment:

$$\mu_2' = E[X^2] = (1/n) \Sigma_{i=1}^{n} x_i^2$$

3. The first central moment:

$$\mu_1 = E[X - \mu]$$

4. The second central moment:

$$\mu_2 = E[(X - \mu)^2]$$

Hence obtain μ, σ^2, and σ for each category, and verify numerically that:

$$\mu_1' = \mu, \quad \mu_1 = 0, \quad \sigma^2 = \mu_2 = \mu_2' - (\mu_1')^2$$

Then, assuming $X \sim N(\mu, \sigma^2)$ for each category, write out the explicit Normal PDF:

$$f(x) = (1 / (\sigma\sqrt{(2\pi)})) \exp[\, -(x - \mu)^2 / (2\sigma^2) \,]$$

for $f_S(x)$, $f_C(x)$, and $f_O(x)$.

Summarize your results in a table with columns: Mean, Second Raw Moment, Variance, Standard Deviation.

Question 5 — Computational Generation and Comparison of PDF Curves

Using the PDFs derived in Question 4, write a Python program that:

1. Defines a **single common x-axis**, e.g. `x = np.linspace(80, 210, 1000)`, used identically for all three curves (same x_{min}, x_{max}, and number of points).
2. Computes $f_S(x)$, $f_C(x)$, $f_O(x)$ over that same grid.
3. Plots **all three PDFs on one graph**, with:
    - x-axis: Top Speed (km/h), y-axis: Probability Density
    - a legend distinguishing Sport, Cruiser, Off-road
    - a suitable title
    - a vertical dashed line marking each category's mean

Graphical interpretation questions:

1. Which two PDF curves appear most similar, and why?
2. Which curve is clearly separated from the other two?
3. Which two curves show the greatest visual overlap?

Question 6 — Covariance and Correlation Between the PDF Curves

The Normal PDF curves for Sport bikes, Cruisers, and Off-road bikes were obtained in Questions 4 and 5. To compare the shapes of these curves numerically, the PDF values were evaluated at the same eight top-speed values.

Use the following table:

Top Speed x (km/h)	Sport PDF S	Cruiser PDF C	Off-road PDF O
90	0.000000	0.000000	0.000255
105	0.000000	0.000000	0.050797

Top Speed x (km/h)	Sport PDF S	Cruiser PDF C	Off-road PDF O
120	0.000000	0.000000	0.013522
135	0.000000	0.000000	0.000005
150	0.000000	0.000000	0.000000
165	0.003107	0.009719	0.000000
180	0.068167	0.050818	0.000000
195	0.001999	0.000032	0.000000

Treat the three PDF columns as the numerical vectors S, C, and O:

$$S = (S_1, S_2, \ldots, S_8), \quad C = (C_1, C_2, \ldots, C_8), \quad O = (O_1, O_2, \ldots, O_8)$$

(a) Calculate the mean PDF value for each vector: $\bar{S}$, $\bar{C}$, and $\bar{O}$.

(b) Using the population covariance formula,

$$\mathrm{Cov}(F, G) = (1/m) \sum_{j=1}^{m} (F_j - \bar{F})(G_j - \bar{G})$$

calculate Cov(S, C), Cov(S, O), and Cov(C, O). Show one covariance calculation completely by hand (for example, Cov(S, C)).

(c) Based on your correlation values and the graph from Question 5:

1. Which two PDF curves are most similar?
2. Which PDF curve is least similar to the other two?
3. Does this agree with the visual comparison from Question 5?
4. Why must all three PDF curves be evaluated at exactly the same x-values when calculating covariance and correlation this way?

Note: the covariance and correlation computed here compare the **sampled PDF values** across a common x-grid; they do *not* measure correlation between the original top-speed random variables.
